has no hair-—the hair being anything that might stick out of the hole to reveal the details of the star from which it was formed.

Box 7.1

An Explanation of Perturbation Methods, for Readers Who Like Algebra

In algebra one learns to compute the square of a sum of two numbers, a and b, from the formula

$$(a + b)^2 = a^2 + 2ab + b^2.$$

Suppose that a is a huge number, for example 1000, and that b is very small by comparison, for example 3. Then the third term in this formula, b^2, will be very small compared to the other two and thus can be thrown away without making much error:

$$(1000+ 3)^2 = 1000^2 + 2 \times 1000 \times 3 + 3^2 = 1{,}006{,}009$$
$$\simeq 1000^2 + 2 \times 1000 \times 3 = 1{,}006{,}000.$$

Perturbation methods are based on this approximation. The $a = 1000$ is like a precisely spherical star, $b = 3$ is like the star's tiny mountain, and $(a + b)^2$ is like the spacetime curvature produced by the star and mountain together. In computing that curvature, perturbation methods keep only effects that are linear in the mountain's properties (effects like $2ab = 6000$, which is linear in $b = 3$); these methods throw away all other effects of the mountain (effects like $b^2 = 9$). So long as the mountain remains small compared to the star, perturbation methods are highly accurate. However, if the mountain were to grow as big as the rest of the star (as it would need to do to make the star square rather than round), then perturbation methods would produce serious errors—errors like those in the above formulas with $a = 1000$ and $b = 1000$:

$$(1000+1000)^2 = 1000^2 + 2 \times 1000 \times 1000 + 1000^2 = 4{,}000{,}000$$